JABS cheatsheet: label and train

1. Get in

- Remote computer: **Windows App** on a Mac, **Remote Desktop (RDP)** on Windows.
- Open the practice project in the JABS GUI: `File → Open Project`, then `data/zenodo_scrubbed/jabs_grooming/`.
- Features are already computed, so the first Train starts quickly.

2. The loop (active learning)

Label → **Train** → **Classify** → **Find mistakes** → **Label them**

Repeat. Each round spends your time on the frames that teach the classifier the most. Always **Classify** after you Train: it writes predictions for every video.

3. How to label

- Label the **confident core** of each bout. **Avoid the edges** (first and last frames): that is where people disagree most.
- **Not behavior** is a label you make on purpose, not whatever is left over.
- A handful of clear bouts of each class is enough to start.
- Sparse labels train the classifier. Dense labels (every frame reviewed) are for checking it.

4. Keyboard shortcuts

Labeling	
`z` `x` `c`	Start select mode. Then `z` = behavior, `c` = not behavior, `x` = clear labels.
`Esc`	Leave select mode without a change.
`Ctrl`+`A`	Select all frames (⌘A on a Mac).
`Ctrl`+`Shift`+`A`	Select the current bout.
Moving around	
`Space`	Play / pause. `Shift`+`Space` plays the current bout.
`←` `→`	Previous / next frame.
`↑` `↓`	Back / forward 10 frames.
`,` `.`	Previous / next video.
`Shift`+`↑`/`↓`	Next / previous subject.
Overlays, search, export	
`p` `l` `t`	Toggle pose (keypoints), landmarks, track overlays.
`Ctrl`+`F`	Behavior search (⌘F on a Mac).
`Ctrl`+`T`	Export training data (⌘T on a Mac).

5. Menus and views

- **View**: *Overlay Pose* shows the keypoints on the mouse (check the pose fits). *Overlay Segmentation* shows each animal's contour (needs a pose file with segmentation). Also *Label Overlay*, *Overlay Landmarks*, *Timeline*.
- **Tools**: *Search Behaviors* and *Prediction Postprocessing*.
- **Features**: switch feature groups on or off (CM Units, Window, Social, Corners, Segmentation).
- Zoom into the video frame and reset it with the restore button. Brightness and contrast controls help when the behavior is small or the lighting is poor.

6. Find frames worth labeling

`Tools → Search Behaviors` (`Ctrl`+`F`) jumps to frames that match a query:

- **label**: where someone labeled the behavior,
- **prediction**: choose **low confidence** to find frames the classifier is least sure about. These are the best to label next,
- **timeline annotation** tag.

7. Train and Classify buttons

- **Classifier type**: choose **XGBoost** (the course default; the GUI starts on Random Forest).
- **Cross Validation k** slider: how many held-out rounds to run. It starts at 1, one noisy fold. Tick **All k-fold** to see the spread across videos.
- **Train** turns on at k = 0 with 20+ labeled frames of behavior and 20+ of not-behavior. With k above 0 you need at least k groups, each with 20+ of both (and never fewer than 2).
- **Classify** predicts every video in the project. It stays off until you have trained.
- Training writes a **training report** in `jabs/training_logs/`.

8. Colors (GUI timeline and every plot)

behavior present · not behavior · unlabeled

JABS cheatsheet: read the results

9. Two ideas to start from

- **Ethogram**: a timeline of behavior. Time runs left to right, one row per video (or annotator, or processing stage), and the color at each moment says what the animal was doing.
- **Window**: the stretch of frames before and after the current frame in which features are calculated. Behavior is temporal, so one frame is not enough. This course uses a window of 60. The idea of window features comes from Kristin Branson's JAABA (Kabra et al. 2013).

10. Metrics

Metric	Asks
Accuracy	What fraction of all frames did we get right? Misleading when the behavior is rare.
Recall	Of all grooming frames, what fraction did we catch?
Precision	Of the frames we called grooming, what fraction were?
F1	Harmonic mean of precision and recall.
AUROC	Area under the ROC curve. Does not depend on the cutoff.
5% FPR at 0.nn TPR	When only 5% of not-grooming frames may be wrongly called grooming, the classifier catches 0.nn of the grooming frames.

ROC curve. Every frame gets a probability of grooming. A call means that probability is above a cutoff (0.5 by default). The curve moves the cutoff and plots the true positive rate (up) against the false positive rate (right). Top-left is better; the diagonal is guessing.

11. Cross-validation: what to hold out

Held out	Score estimates performance on…
Video	a new recording
Animal	a new animal (needs an animal ID in the filename, or one animal per video)
Filename pattern	a new arena or cage (needs the ID in the filename)

The stricter the grouping, the more honest and the noisier the score. Read the **spread** across folds in the training report, not just the mean.

12. From frames to bouts

- **Stitch**: fill a short gap of not-behavior inside a behavior bout.
- **Filter**: drop a predicted bout that is too short to be real.
- Compare classifiers on bouts, not just on frames. *Tools → Prediction Postprocessing* in the GUI.

13. Next step: new data (introduced, not run today)

Apply the classifier you built to videos nobody labeled:

```
jabs-classify classify --classifier <classifier> --input-pose <pose file> --out-dir <folder>
```

then `jabs-cli postprocess` for filter + stitch. Validate on your own ground truth before trusting a classifier, yours or someone else's.

14. Links

- Course repository: github.com/KumarLabJax/getting-started-with-jabs
- JABS documentation: kumarlabjax.github.io/JABS-behavior-classifier (user guide, keyboard shortcuts, CLI)
- JABS GUI walkthrough: kumarlabjax.github.io/getting-started-with-jabs
- Grooming example projects (Zenodo): doi:10.5281/zenodo.16697331
- Aging dataset (EMBER-DANDI): doi:10.60533/ember-dandi.000466/0.261009.2220
- BXD dataset (Harvard Dataverse): doi:10.7910/DVN/RQYI04
- Questions about JABS: jabs@jax.org

JABS paper: Choudhary et al., eLife 2026, elifesciences.org/articles/107259. Window features: Kabra et al., Nature Methods 2013, doi:10.1038/nmeth.2281.
Keyboard shortcuts and menus follow the JABS user guide.